

Conjecture on the Number of Linearly Independent Relations of Ihara–Kaneko–Zagier Derivation Relation: An Euler-type Pentagonal Formula

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1. GENERALIZED PENTAGONAL NUMBERS AND THE $\omega(n)$ FUNCTION

Let $P_0, P_1, P_2, P_3, P_4, P_5, P_6, P_7, \dots = 0, 1, 2, 5, 7, 12, 15, 22, \dots$ denote the generalized pentagonal numbers OEIS:A001318, and let $\omega(k) = A010815(k)$, defined by

$$\omega(k) = \begin{cases} 1 & \text{if } k = 0; \\ (-1)^m & \text{if } k = \frac{3m^2 \pm m}{2}; \\ 0 & \text{otherwise.} \end{cases}$$

Table 1. P_n and $\omega(n)$

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
P_n	0	1	2	5	7	12	15	22	26	35	40	51	57	70	77	92	100	117	126	145	155	176	187	210
$\omega(n)$	1	-1	-1	0	0	1	0	1	0	0	0	0	-1	0	0	-1	0	0	0	0	0	0	1	0

2. THE CONJECTURE AND SOME EQUIVALENT FORMULATIONS

For a given weight $n \geq 3$ let $\mathcal{L}(n)$ denote the the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation. In [2, 3] the authors have tabulated the values of $\mathcal{L}(n)$ calculated using the Risa/Asir and Mathematica general computer algebra system for $n = 3$ to $n = 15$.

Table 2. Number of Independent Derivation Relations for MZVs

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16*	17*
$\mathcal{L}(n)$	1	2	5	10	22	44	90	181	363	727	1456	2912	5825	11652*	23304*

* Calculated with a JavaScript application of my own development.¹

Conjecture 1. *For all weight $n \geq 3$ the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation can be calculated by the following Euler-type pentagonal formula:*

$$\mathcal{L}(n) = 2^{n-2-P_1} + 2^{n-2-P_2} - 2^{n-2-P_3} - 2^{n-2-P_4} \pm \dots - \omega(n-1) - 1 = - \sum_{k=1}^{n-2} \omega(k) 2^{n-2-k} - \omega(n-1) - 1.$$

The generating function of $\mathcal{L}(n)$ is given by:

$$F(x) = \frac{x^2}{x-1} + \frac{x(x-1)}{2x-1} \prod_{k \geq 1} (1-x^k),$$

and $\mathcal{L}(n)$ satisfies the following recurrence relation for all $n \geq 3$:

$$\mathcal{L}(n+1) = 2 \cdot \mathcal{L}(n) + 1 - \omega(n) + \omega(n-1).$$

The calculations below verify that Conjecture 1. holds for $n = 3$ through $n = 17$:

$$\mathcal{L}(3) = 2^{1-1} - \omega(2) - 1 = 2^0 + 1 - 1 = 1 + 1 - 1 = 1$$

$$\mathcal{L}(4) = 2^{2-1} + 2^{2-2} - \omega(3) - 1 = 2^1 + 2^0 - 0 - 1 = 2 + 1 - 1 = 2$$

$$\mathcal{L}(5) = 2^{3-1} + 2^{3-2} - \omega(4) - 1 = 2^2 + 2^1 - 0 - 1 = 4 + 2 - 1 = 5$$

$$\mathcal{L}(6) = 2^{4-1} + 2^{4-2} - \omega(5) - 1 = 2^3 + 2^2 - 1 - 1 = 8 + 4 - 1 - 1 = 10$$

$$\mathcal{L}(7) = 2^{5-1} + 2^{5-2} - 2^{5-5} - \omega(6) - 1 = 2^4 + 2^3 - 2^0 - 0 - 1 = 16 + 8 - 1 - 1 = 22$$

¹ On page <https://tamas-tech.github.io>

$$\begin{aligned}
\mathcal{L}(8) &= 2^{6-1} + 2^{6-2} - 2^{6-5} - \omega(7) - 1 = 2^5 + 2^4 - 2^1 - 1 - 1 = 32 + 16 - 2 - 1 - 1 = 44 \\
\mathcal{L}(9) &= 2^{7-1} + 2^{7-2} - 2^{7-5} - 2^{7-7} - \omega(8) - 1 = 2^6 + 2^5 - 2^2 - 2^0 - 0 - 1 = 64 + 32 - 4 - 1 - 1 = 90 \\
\mathcal{L}(10) &= 2^{8-1} + 2^{8-2} - 2^{8-5} - 2^{8-7} - \omega(9) - 1 = 2^7 + 2^6 - 2^3 - 2^1 - 0 - 1 = 128 + 64 - 8 - 2 - 1 = 181 \\
\mathcal{L}(11) &= 2^{9-1} + 2^{9-2} - 2^{9-5} - 2^{9-7} - \omega(10) - 1 = 2^8 + 2^7 - 2^4 - 2^2 - 0 - 1 = 256 + 128 - 16 - 4 - 1 = 363 \\
\mathcal{L}(12) &= 2^{10-1} + 2^{10-2} - 2^{10-5} - 2^{10-7} - \omega(11) - 1 = 2^9 + 2^8 - 2^5 - 2^3 - 0 - 1 = 512 + 256 - 32 - 8 - 1 = 727 \\
\mathcal{L}(13) &= 2^{11-1} + 2^{11-2} - 2^{11-5} - 2^{11-7} - \omega(12) - 1 = 2^{10} + 2^9 - 2^6 - 2^4 + 1 - 1 = 1024 + 512 - 64 - 16 = 1456 \\
\mathcal{L}(14) &= 2^{12-1} + 2^{12-2} - 2^{12-5} - 2^{12-7} + 2^{12-12} - \omega(13) - 1 = 2^{11} + 2^{10} - 2^7 - 2^5 + 2^0 - 0 - 1 = 2048 + 1024 - 128 - 32 + 1 - 1 = 2912 \\
\mathcal{L}(15) &= 2^{13-1} + 2^{13-2} - 2^{13-5} - 2^{13-7} + 2^{13-12} - \omega(14) - 1 = 2^{12} + 2^{11} - 2^8 - 2^6 + 2^1 - 0 - 1 = 4096 + 2048 - 256 - 64 + 2 - 1 = 5825 \\
\mathcal{L}(16) &= 2^{14-1} + 2^{14-2} - 2^{14-5} - 2^{14-7} + 2^{14-12} - \omega(15) - 1 = 2^{13} + 2^{12} - 2^9 - 2^7 + 2^2 + 1 - 1 = 8192 + 4096 - 512 - 128 + 4 = 11652 \\
\mathcal{L}(17) &= 2^{15-1} + 2^{15-2} - 2^{15-5} - 2^{15-7} + 2^{15-12} + 2^{15-15} - \omega(16) - 1 = 2^{14} + 2^{13} - 2^{10} - 2^8 + 2^3 + 2^0 - 0 - 1 = \\
&= 16384 + 8192 - 1024 - 256 + 8 + 1 - 1 = 23304
\end{aligned}$$

The first equivalent version reveals a simple connection between $\mathcal{L}(n)$ and $\text{pic}(n) = \text{A178841}(n)$ the number of pure inverting compositions of n .

Conjecture 2. For all weight $n \geq 3$ the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation can be calculated by the following formula:

$$\mathcal{L}(n) = 2^{n-2} - 1 - \text{pic}(n-1)$$

The calculations summarized in the table below verify that Conjecture 2. holds for $n = 3$ through $n = 17$:

Table 3. $\mathcal{L}(n) = 2^{n-2} - 1 - \text{pic}(n-1)$

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$2^{n-2} - 1$	1	3	7	15	31	63	127	255	511	1023	2047	4095	8191	16383	32767
$\text{pic}(n-1)$	0	1	2	5	9	19	37	74	148	296	591	1183	2366	4731	9463
$2^{n-2} - 1 - \text{pic}(n-1)$	1	2	5	10	22	44	90	181	363	727	1456	2912	5825	11652	23304

Let $c(n) = \text{A152537}(n)$. The second equivalent reformulation highlights a deeper algebraic relationship between $c(n)$ and $\text{pic}(n)$.

Conjecture 3. For all weight $n \geq 3$ the number of linearly independent relations of Ihara-Kaneko-Zagier derivation relation can be calculated by the following formula:

$$\mathcal{L}(n) = 2^{n-2} - 1 - \omega(n-1) - c(n-2)$$

The calculations summarized in the table below verify that Conjecture 3. holds for $n = 3$ through $n = 17$:

Table 4. $\mathcal{L}(n) = 2^{n-2} - 1 - \omega(n-1) - c(n-2)$

n	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$2^{n-2} - 1$	1	3	7	15	31	63	127	255	511	1023	2047	4095	8191	16383	32767
$\omega(n-1)$	-1	0	0	1	0	1	0	0	0	0	-1	0	0	-1	0
$c(n-2)$	1	1	2	4	9	18	37	74	148	296	592	1183	2366	4732	9463
$2^{n-2} - 1 - \omega(n-1) - c(n-2)$	1	2	5	10	22	44	90	181	363	727	1456	2912	5825	11652	23304

3. EMPIRICAL AND THEORETICAL RESULTS LEADING TO CONJECTURE 1

Before proceeding, we establish some terminology and notation that will be used throughout the remainder of this work.

Let $\mathfrak{H}_n := \{w \in \mathfrak{H}, |w| = n\}$ denote the set of words of weight n . We recall that the set of admissible words is given by $\mathcal{A} = x\mathfrak{H}y$, and we denote the subset of admissible words of weight n by $\mathcal{A}_n := x\mathfrak{H}_{n-2}y$. For any admissible word $w \in \mathcal{A}$, we define its binary value, denoted by $\text{bin}(w)$, as the unique even natural number whose binary representation yields the word w upon replacing each digit 1 with x and each digit 0 with y . Equivalently, if a word is written as $w = u_1u_2 \dots u_k$, where $u_i \in \{x, y\}$ then

$$\text{bin}(w) = \sum_{i=1}^k c_i 2^{k-i}, \text{ where } c_i = 1 \text{ if } u_i = x, \text{ and } c_i = 0 \text{ if } u_i = y.$$

For example, consider the admissible word $w = xxyy$. Replacing each x with 1 and y with 0 gives the binary string 1100₂. Converting this to base 10, we obtain: $\text{bin}(xxyy) = 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 0 \cdot 2^0 = 8 + 4 = 12$.

We will use a graded lexicographic total order on the words of $\mathfrak{H}$, defined as follows. If u and v are words with $|u| < |v|$, set $u < v$. If $|u| = |v|$ and $u = wyw_2$ and $v = wxw_3$, for any words w, w_1 , and w_2 , set $u < v$. It is easy to see that the order on $\mathcal{A}$ induced by the mapping $\text{bin}(\cdot)$ coincides with the graded lexicographic order restricted to $\mathcal{A}$. The words of degree at most 3 are thus ordered as follows:

$$1 < y < x < yy < yx < xy < xx < yyy < yyx < yxy < yxx < xyy < xyx < xxy < xxx$$

For any subset $V \subseteq \mathfrak{H}$ and any integer $m \geq 1$, let $\partial_m(V)$ denote the set $\{\partial_m(w) : w \in V\}$.

Conjecture 1 is based on the following analysis of the rank growth, which chronologically preceded the formulation of Conjecture 2.

For a fixed $n \geq 3$, we will construct a full row-rank matrix $\mathbf{M}_n$ from the derivatives $\partial_{n-|w|}(w)$ of the elements w in the subset

$$\mathcal{B}_n = \mathcal{A}_{n-1} \cup x\mathcal{A}_{n-3} \cup x\mathcal{A}_{n-2} \cup \cdots \cup x\mathcal{A}_2 \cup \mathcal{A}_2 \subset h$$

($|\mathcal{B}_n| = 2^{n-3} + (2^{n-5} + 2^{n-6} + \cdots + 2^0) + 1 = 3 \cdot 2^{n-4}$) according to the following procedure:

The columns of $\mathbf{M}_n$ are indexed by the elements of the set of admissible words of $\mathcal{A}_n$, ordered ascendingly with respect to the graded lexicographic order.

- (1) Let $w = x^{n-2}y$ be the maximal element of the set $\mathcal{A}_{n-1}$. The first row of the matrix $\mathbf{M}_n$ is defined as the coordinate vector of the derivative $\partial_1(w) \in \partial_1(\mathcal{A}_{n-1}) \subset \text{Span}(\mathcal{A}_n)$ relative to the ordered basis $(\mathcal{A}_n, <)$.
- (2) We then proceed iteratively through the remaining elements of $\mathcal{A}_{n-1}$ in descending order. For each element, we compute the coordinate vector of its derivative and insert it as a new row in $\mathbf{M}_n$. *Crucially, throughout this process, we ensure that if a candidate row can be expressed as a linear combination of the preceding rows, it is discarded.*
- (3) This procedure is then repeated analogously for the elements of the remaining subsets in $\mathcal{B}_n$ (namely $x\mathcal{A}_{n-3}, x\mathcal{A}_{n-2}, \dots, x\mathcal{A}_2$, and $\mathcal{A}_2$ in the given order), examining their corresponding derivatives in descending order within each subset.

Constructing the matrix via this elimination process, it is evident that the resulting matrix $\mathbf{M}_n$ will have full row-rank. At the end of the procedure, we count how many derivatives originating from the subsets $\mathcal{A}_{n-1}, x\mathcal{A}_{n-3}, x\mathcal{A}_{n-2}, \dots, x\mathcal{A}_2$, and $\mathcal{A}_2$ remain as rows in the matrix. These quantities will be denoted by $r_{0,n-2}, r_{1,n-2}, r_{2,n-2}, \dots, r_{n-3,n-2}$ respectively, and they will constitute the entries of the $(n-2)$ -th column of an upper triangular matrix $\mathbf{R}$, whose rows and columns are indexed starting from 0.

Since our JavaScript application, which includes the implementation of the above procedure, generates a clear graphical layout for smaller values of n , the visual output representing the matrix $\mathbf{M}_6$ for $n = 6$ is included below as an illustrative example.

$\mu(w)$	ρ	D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
w	rank	Delete	xyyyy	xyyyxy	xyxyxy	xyyxyx	xyxyyy	xyxyxy	xyxxxy	xyxxxxy	xyyyy	xxxyxy	xxxyxy	xxxyxy	xxxyxy	xxxyxy	xxxyxy	xxxyxy
$\partial_1^1(\text{xxxxxy})$	1	×	0	0	0	0	0	0	0	1	0	0	0	1	0	1	1	-1
$\partial_1^1(\text{xxxxyy})$	2	×	0	0	0	0	0	0	1	0	0	0	1	0	1	-1	-1	0
$\partial_1^1(\text{xxyxyx})$	3	×	0	0	0	0	0	1	0	0	0	1	1	-1	0	-1	0	0
$\partial_1^1(\text{xxyyyx})$	4	×	0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0
$\partial_1^1(\text{xyxxxy})$	5	×	0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0
$\partial_1^1(\text{xyxyxy})$	6	×	0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0	0
$\partial_1^1(\text{xyyyxy})$	7	×	0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0
$\partial_1^1(\text{xyyyyy})$	8	×	1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0
$\partial_2^1(\text{xxxxy})$	9	×	0	0	0	1	0	0	0	0	0	1	0	1	1	1	0	-1
$\partial_2^1(\text{xxyyy})$	10	×	0	0	1	0	0	0	0	0	1	0	0	-1	0	0	-1	0

For $n = 6$, the generating set is given by $\mathcal{B}_6 = \mathcal{A}_5 \cup x\mathcal{A}_3 \cup x\mathcal{A}_2 \cup \mathcal{A}_2$. As clearly observed from the visual output, all $2^3 = 8$ elements of $\partial_1(\mathcal{A}_5)$ and all $2^1 = 2$ elements of $\partial_2(x\mathcal{A}_3)$ remain in the matrix. Conversely, the elements of $\partial_3(x\mathcal{A}_2) = \{\partial_3(xxy)\}$

To provide an additional illustration, let us consider the output generated for the case $n = 7$.

$\mu(w)$	p	D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
w	rank	Label	xxxxxyy	xyyyxyx	xyyxxyy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy	xyjyxxy
$\partial_1^1(\text{xxxxxy})$	1		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0	1	1	-1
$\partial_1^1(\text{xxxxxy})$	2		0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0	1	-1	-1	0
$\partial_1^1(\text{xxxxyx})$	3		0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	1	-1	0	-1	0	0
$\partial_1^1(\text{xxxxyy})$	4		0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0
$\partial_1^1(\text{xyyxyx})$	5		0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0
$\partial_1^1(\text{xyyxyy})$	6		0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0	0
$\partial_1^1(\text{xyyyxy})$	7		0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0
$\partial_1^1(\text{xyyyyy})$	8		0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0
$\partial_1^1(\text{xyxxxy})$	9		0	0	0	0	0	0	0	1	0	0	0	1	0	1	1	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0
$\partial_1^1(\text{xyxxxxy})$	10		0	0	0	0	0	0	1	0	0	0	1	0	1	-1	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0
$\partial_1^1(\text{xyxyxy})$	11		0	0	0	0	0	1	0	0	0	1	1	-1	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0
$\partial_1^1(\text{xyxyyy})$	12		0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0
$\partial_1^1(\text{xyyxyx})$	13		0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_1^1(\text{xyyxyy})$	14		0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_1^1(\text{xyyyxy})$	15		0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_1^1(\text{xyyyyy})$	16		1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_2^1(\text{xxxxxy})$	17		0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	1	1	1	0	-1
$\partial_2^1(\text{xxxxxy})$	18		0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	0	-1	0	0	-1	0
$\partial_2^1(\text{xxxxyx})$	19		0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	-1	0	0	0	0	0	-1	0	0
$\partial_2^1(\text{xxxxyy})$	20		0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	-1	-1	0	0	-1	0	0	0	0	0	-1	0	0	0	0
$\partial_3^1(\text{xxxxy})$	21		0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	1	0	0	1	1	1	1	0	0	0	-1
$\partial_3^1(\text{xxxxy})$	22		0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0	1	0	0	-1	-1	-1	0	0	0	0	0	0	0	-1	0

From this output, it can be observed that all 16, 4, and 2 elements of $\partial_1(\mathcal{A}_6)$, $\partial_2(x\mathcal{A}_4)$, and $\partial_3(x\mathcal{A}_3)$, respectively, remain in the matrix, whereas all elements of $\partial_4(x\mathcal{A}_2)$ and $\partial_5(\mathcal{A}_2)$ are eliminated. By definition, this yields $r_{0,5} = 16$, $r_{1,5} = 4$, $r_{2,5} = 2$, and $r_{3,5} = r_{4,5} = 0$.

Next, we present the empirical results obtained for $r_{i,j}$ from $n = 2$ to $n = 17$, collected into the table below:

[illegible]

The most significant observation regarding the resulting table is that, putting aside the first few entries of each row, the values follow a clearly discernible recurrence relation. Specifically, the j -th entry of the i -th row can be obtained as the difference between the $(j-1)$ -th entry of the $(i-1)$ -th row and the $(j-i-1)$ -th entry of the $(i-1)$ -th row.

Formally, for sufficiently large indices, the entries satisfy the following recurrence: $r_{i,j} = r_{i-1,j-1} - r_{i-1,j-i-1}$. To illustrate this pattern, consider the following examples taken directly from the empirical data:

- For $i = 4$ and $j = 12$ (corresponding to ∂_5 at $n = 14$), we have $39 = 42 - 3$.
- For $i = 3$ and $j = 13$ (corresponding to ∂_4 at $n = 15$), we have $168 = 192 - 24$.

By applying this recurrence relation backward, we can determine the exact initial values required at the beginning of each row for the relation to hold globally across the entire table. Although such a backward extension could potentially lead to a contradiction, remarkably, no such inconsistency arises. Specifically, if we set all entries on the main diagonal to 1 and all entries strictly below the main diagonal to 0, the recurrence relation is satisfied consistently throughout the entire table. The resulting extended table is presented below:

$i \setminus j$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	1	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	...
1	0	1	0	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	...
2	0	0	1	0	0	2	3	6	12	24	48	96	192	384	768	1536	...
3	0	0	0	1	0	0	1	3	6	10	21	42	84	168	336	672	...
4	0	0	0	0	1	0	0	1	2	6	10	20	39	78	158	315	...
5	0	0	0	0	0	1	0	0	1	2	5	10	20	38	76	152	...
6	0	0	0	0	0	0	1	0	0	1	2	5	9	20	38	75	...
7	0	0	0	0	0	0	0	1	0	0	1	2	5	9	19	38	...
8	0	0	0	0	0	0	0	0	1	0	0	1	2	5	9	19	...
9	0	0	0	0	0	0	0	0	0	1	0	0	1	2	5	9	...
10	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	5	...
11	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	2	...
12	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	...
13	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	...
14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	...
15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$
$\sum_{j=1}^i a_{i,j}$	1	2	3	6	11	23	45	91	182	364	728	1457	2913	5826	11653	23305	...

It naturally follows from the construction of the table that subtracting 1 from the sum of the entries in its j -th column yields exactly the values of $\mathcal{L}(j+2)$ for all $j \geq 1$.

A formalization of the recurrence relation observed in the table is given by:

$$r_{i,j} = \begin{cases} 0 & \text{if } i < 0 \text{ or } j < 0; \\ 1 & \text{if } i = j \text{ for } i = 0, 1, 2, 3, \dots; \\ 2^{\max(0, j-1)} & \text{if } i = 1 \text{ for } j = 0, 1, 2, 3, \dots; \\ a_{i-1, j-1} - a_{i-1, j-i-1} & \text{otherwise.} \end{cases}$$

The recurrence relation was investigated within a more general theoretical framework as a transformation. Instead of the initial values $2^{\max(0, j-1)}$ entered into the first row, let us consider an arbitrary arithmetic sequence $\mathbf{a} = (a(1), a(2), \dots)$, and let the column sums define the values of the transformed sequence $T(\mathbf{a}) = (T(a(1)), T(a(2)), \dots)$. During our thorough theoretical investigation, it soon became apparent that this transformation is remarkably versatile and well-suited for providing concise and clear explanations of the connections between various combinatorial results with a special emphasis on pentagonal formulas. Furthermore, since it is based on shifted differences and a summation, the transformation is inherently linear. What is essential for our purposes is that the transformation maps the sequence $\mathbf{a} = (1, 0, 0, 0, \dots)$ to the sequence $-\boldsymbol{\omega} = (1, 1, 0, 0, -1, 0, -1, 0, 0, 0, 0, 1, 0, \dots)$ (now, starting from 1 rather than 0 as done above). Consequently, by linearity, the transform of an arbitrary sequence $\mathbf{a} = (a_1, a_2, a_3, \dots)$ yields a pentagonal formula of the form $-\sum_{k=1}^{\infty} \omega(k) a_{n-k}$, which can be written more concisely as $T(\mathbf{a}) = -(a_1, a_2, \dots) * (\omega(1), \omega(2), \dots)$, where $*$ denotes the convolution of sequences. This, along with a few straightforward algebraic manipulations, directly recovers the pentagonal formula presented in Conjecture 1. Since the specific areas discussing this type of transformation lie beyond the current scope of our knowledge, we shall adopt the term *pentagonal transformation* for the remainder of this work. During the study of the transformation, it also became apparent that when transforming various sequences $\mathbf{a} = (a(1), a(2), \dots)$, it is not only the transformed sequence $T(\mathbf{a})$ itself that is of interest; the individual rows generated in the table during the underlying recurrence relation, as well as their corresponding generating functions, are also highly significant. In the table above leading to the values of $\mathcal{L}(j)$, for instance, it is clearly observable that

the first non-zero entries of successive rows coincide with the sequence $(\text{pic}(0), \text{pic}(1), \text{pic}(2), \dots) = (1, 0, 0, 1, 2, 5, 9, 19, 37, \dots)$ for increasingly longer stretches. From this, together with certain relations to be discussed in Section 5, it will become completely evident that each row of the table contains parametric generalizations $(\text{pic}_n(0), \text{pic}_n(1), \text{pic}_n(2), \dots)$ of the sequence $(\text{pic}(0), \text{pic}(1), \text{pic}(2), \dots)$ corresponding to the parameter n of that row.

As for the proof of Conjecture 1, it suffices to demonstrate two conditions:

- (1) for any $n \geq 3$, the vectors in the set $\partial_1(\mathcal{A}_{n-1})$ are linearly independent;
- (2) the rank increments $r_{i,j}$ follow the aforementioned recurrence relation.

The first statement can be straightforwardly established without any difficulty, from which the corresponding power of two naturally follows. However, we have been unable to provide a proof for the second statement.

It is worth emphasizing that the credibility of Conjecture 1 is significantly reinforced by the fact that the consistency does not merely rest on the first fourteen terms of a single sequence. Rather, it is deeply embedded in the entire underlying table of rank increments, matching perfectly across the whole two-dimensional recursive structure.

4. EMPIRICAL AND THEORETICAL RESULTS LEADING TO CONJECTURE 2

The empirical results leading to Conjecture 2 can be obtained via the exact same elimination procedure described for Conjecture 1; however, rather than traversing the words in descending order, the matrix is iteratively expanded by examining the coordinate vectors of their derivatives in ascending order. This modification offers a distinct computational advantage: as can be readily verified, the coordinate vectors of the derivatives in the set

$$\mathcal{B}_n^* = \mathcal{B}_n \setminus \mathcal{A}_{n-1} = \partial_{n-2}(\mathcal{A}_2) \cup \partial_{n-3}(x\mathcal{A}_2) \cup \dots \cup \partial_2(x\mathcal{A}_{n-3})$$

relative to the ordered basis $(\mathcal{A}_n, <)$ are guaranteed to form a linearly independent system of vectors. Consequently, for any fixed $n \geq 3$, the elimination procedure can be initiated immediately with the full row-rank matrix $\mathbf{N}_n$ whose rows are defined as the coordinate vectors of the derivatives contained in the set $\mathcal{B}_n^*$ without requiring any prior rank computation. This initial matrix $\mathbf{N}_n$ consists of exactly $1 + (2^0 + 2^1 + \dots + 2^{n-6} + 2^{n-5}) = 2^{n-4}$ rows. Subsequently, we iterate through the remaining elements of the set $\mathcal{A}_{n-1}$ in ascending order, starting from its minimal element, xy^{n-2} . For each successive word w , we append the coordinate vector of its derivative $\partial_1(w)$ as a candidate row to $\mathbf{N}_n$; if this candidate row increases the rank of the matrix, it is retained, and otherwise, it is discarded. Upon completion of this procedure, the resulting matrix $\mathbf{N}_n$ is inherently full row-rank, just like the matrix $\mathbf{M}_n$ discussed in Conjecture 1. Furthermore, the ranks of these two matrices are identical.

Since our custom JavaScript application—which implements the aforementioned procedure too—generates a clear graphical layout for small values of n , the visual outputs representing the matrices $\mathbf{N}_6$ and $\mathbf{N}_7$ are included below as illustrative examples.

$\mu(w)$	ρ	D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
w	rank	Delete	xyyyyy	xyyyxy	xyxyxy	xyxxxy	xyxyyy	xyxyxy	xyxxxy	xyxxxy	xyxyyy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy
$\partial_4^1(xy)$	1	×	1	0	1	0	1	0	1	0	0	-1	0	-1	0	-1	0	-1
$\partial_3^1(xxy)$	2	×	0	1	0	0	0	1	0	0	1	1	1	0	0	0	0	-1
$\partial_2^1(xxyy)$	3	×	0	0	1	0	0	0	0	0	1	0	0	-1	0	0	-1	0
$\partial_2^1(xxxy)$	4	×	0	0	0	1	0	0	0	0	0	1	0	1	1	1	0	-1
$\partial_1^1(xyyyy)$	5	×	1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0
$\partial_1^1(xyyxy)$	6	×	0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0
$\partial_1^1(xyxyy)$	7	×	0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0	0
$\partial_1^1(xyxxxy)$	8	×	0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0
$\partial_1^1(xxyyy)$	9	×	0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0
$\partial_1^1(xxxyy)$	10	×	0	0	0	0	0	0	1	0	0	0	1	0	1	-1	-1	0

$\mu(w)$	p	D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
w	rank	Delte	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy	xyxyxy
$\partial_3^1(xy)$	1	×	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	0	-1	0	-1	0	-1	0	-1	0	-1	0	-1	0	-1	0	-1
$\partial_4^1(xoxy)$	2	×	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	1	1	1	0	1	1	1	0	0	0	0	-1	0	0	0	-1
$\partial_5^1(xoxyy)$	3	×	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	0	-1	-1	-1	0	0	0	0	0	0	-1	0
$\partial_6^1(xoxyxy)$	4	×	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0	1	0	1	0	0	1	1	1	1	0	0	0	-1	0
$\partial_7^1(xoxyxyy)$	5	×	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	-1	-1	0	0	-1	0	0	0	0	0	-1	0	0	0
$\partial_8^1(xoxyxyxy)$	6	×	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	-1	0	0	0	0	0	-1	0	0
$\partial_9^1(xoxyxyxyy)$	7	×	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	0	0	-1	0	0	-1	0
$\partial_{10}^1(xoxyxyxyxy)$	8	×	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	1	1	1	0	-1
$\partial_1^1(xyxyxyxy)$	9	×	1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_2^1(xyxyxyxy)$	10	×	0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_3^1(xyxyxyxy)$	11	×	0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_4^1(xyxyxyxy)$	12	×	0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0
$\partial_5^1(xyxyxyxy)$	13	×	0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0
$\partial_6^1(xyxyxyxy)$	14	×	0	0	0	0	0	1	0	0	0	1	1	-1	0	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0
$\partial_7^1(xyxyxyxy)$	15	×	0	0	0	0	0	0	1	0	0	0	1	0	1	-1	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0
$\partial_8^1(xyxyxyxy)$	16	×	0	0	0	0	0	0	0	1	0	0	0	1	0	1	1	-1	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0
$\partial_9^1(xyxyxyxy)$	17	×	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	-1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0	0
$\partial_{10}^1(xyxyxyxy)$	18	×	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	-1	0	-1	0	0	0	-1	0	0	0	0	0	0
$\partial_{11}^1(xyxyxyxy)$	19	×	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	1	-1	-1	0	0	0	-1	0	0	0	0
$\partial_{12}^1(xyxyxyxy)$	20	×	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1	1	-1	0	0	0	-1	0	0	0	0
$\partial_{13}^1(xyxyxyxy)$	21	×	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	-1	-1	0	-1	0	0	0
$\partial_{14}^1(xyxyxyxy)$	22	×	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	1	-1	0	-1	0	0

Although it is theoretically possible that a word $w_1 \in \mathcal{A}_{n-1}$ is skipped while the derivative of a larger word $w_2 > w_1$ is retained in the matrix $\mathbf{N}_n$, empirical evidence suggests that this never occurs for $n > 6$. Specifically, for any weight $n \neq 6$, there exists a unique critical threshold word $w_{\text{crit}} \in \mathcal{A}_{n-1}$ such that its derivative is discarded, and consequently, the derivatives of all subsequent words $w > w_{\text{crit}}$ are also eliminated. In short, with the sole exception of $n = 6$, elements are systematically discarded exclusively from the tail of the ordered set $\mathcal{A}_{n-1}$. (The anomalous behavior observed in the case $n = 6$ is explicitly accounted for by the "congestion" of the pentagonal numbers 3 and 5.)

In light of these observations, *Conjecture 2 can be stated in a sharper form as follows*: for any weight $n \neq 6$, during the construction of the matrix $\mathbf{N}_n$, the first and minimal word $w_0 \in \mathcal{A}_{n-1}$ whose derivative fails to increase the rank of the matrix is precisely the $(3 \cdot 2^{n-4} - \text{pic}(n-1))$ -th element in the ordered set. Consequently, the derivatives of all words larger than w_0 likewise fail to increment the rank of the matrix.

Since $|\mathcal{B}_n^*| = 2^{n-4}$, and exactly $3 \cdot 2^{n-4} - \text{pic}(n-1) - 1$ elements from $\mathcal{A}_{n-1}$ are retained in the matrix $\mathbf{N}_n$, it follows that:

$$\mathcal{L}(n) = \text{rank}(\mathbf{N}_n) = 2^{n-4} + 3 \cdot 2^{n-4} - \text{pic}(n-1) - 1 = 2^{n-2} - \text{pic}(n-1) - 1,$$

which precisely recovers the formulation of Conjecture 2 presented in Section 2. Although the equivalence between Conjecture 1 and Conjecture 2 can be formally established with ease in light of the identities to be discussed in Section 5 (where we shall demonstrate that the rank increments are fundamentally parametric generalizations of pure inverting compositions), we consider it essential to present these two approaches separately, as the structural nuances arising between them are not fully captured by the explicit formulas alone.

It is worth noting that this particular formulation and matrix construction are highly optimized for computationally efficient implementations, which could potentially enable the verification of the conjecture for weights well beyond $n = 17$.

It is highly instructive to reflect upon a comment by N. J. A. Sloane in the OEIS entry for **A178841**, where he noted that the striking similarity between the terms of **A178841**(n) and **A152537**(n) might be "just a coincidence, caused by the fact that their generating functions are almost identical." Our framework provides a rigorous, deeper theoretical explanation that dispels this assumption of coincidence. The connection between these two sequences is fundamentally rooted in the dual matrix constructions presented herein. Specifically, building the derivation matrix in descending order yields a rank-increment structure governed by **A152537** (Conjecture 1), whereas building it in ascending order yields a structure governed by **A178841** (Conjecture 2). Since

both procedures ultimately resolve the exact same system of Ihara-Kaneko-Zagier derivation relations, the ranks of the resulting matrices $\mathbf{M}_n$ and $\mathbf{N}_n$ must identically coincide with $\mathcal{L}(n)$, thereby intrinsically forcing the near-identity of their respective generating functions.

5. GENERALIZATION OF $\text{PIC}(n)$

Preliminary draft.

In this section, we demonstrate that the rank increments can be viewed as a natural generalization of the classical sequence $\text{pic}(k)$ to the parametric function $\text{pic}_n(k)$.

Before presenting the formal identities, we define three fundamental families of polynomials that govern these combinatorial structures. Let Fib_n and Fab_n denote the multivariate Fibonacci and Faber partition polynomials, respectively. Furthermore, let Z_n denote the cycle index polynomial of the symmetric group S_n . For a given set of variables $(x_1, x_2, \dots, x_n)$, these polynomials are explicitly defined as follows:

$$\begin{aligned} \text{Fib}_n(x_1, x_2, \dots, x_n) &= \sum_{1b_1+2b_2+\dots+nb_n=n} (-1)^{b_1+\dots+b_n} (b_1+\dots+b_n)! \frac{x_1^{b_1} x_2^{b_2} \dots x_n^{b_n}}{b_1! b_2! \dots b_n!} \\ \text{Fab}_n(x_1, x_2, \dots, x_n) &= \sum_{1b_1+2b_2+\dots+nb_n=n} (-1)^{b_1+\dots+b_n} n (b_1+\dots+b_n-1)! \frac{x_1^{b_1} x_2^{b_2} \dots x_n^{b_n}}{b_1! b_2! \dots b_n!} \\ Z_n(x_1, x_2, \dots, x_n) &= \sum_{1b_1+2b_2+\dots+nb_n=n} \frac{1}{1^{b_1} 2^{b_2} \dots n^{b_n}} \frac{x_1^{b_1} x_2^{b_2} \dots x_n^{b_n}}{b_1! b_2! \dots b_n!} \end{aligned}$$

Motivated by the theoretical results established in [1], we present the fundamental recurrence relation satisfied by $\text{pic}(k)$, along with several formulas that characterize the intricate relations among the pure inverting composition function $\text{pic}(k)$, the partition function $p(k)$, and the divisor sum function $\sigma(k)$.

$$\begin{aligned} \text{pic}(k) &= \sum_{j=0}^k \omega(j) 2^{k-1-j} + \omega(j) \\ k \text{ pic}(k) &= \sum_{j=0}^k \text{pic}(k-j) (2^j - \sigma(j) - 1) \\ \text{pic}(k) &= 2 \text{ pic}(k-1) + \omega(k) - \omega(k-1) \\ \text{pic}(k) &= Z_k(2^1 - \sigma(1) - 1, 2^2 - \sigma(2) - 1, \dots, 2^k - \sigma(k) - 1) \\ \sigma(k) - 2^k + 1 &= \text{Fab}_k(\text{pic}(1), \text{pic}(2), \dots, \text{pic}(k)) \\ \sum_{j=0}^k p(j) - 2p(k) &= -\text{Fib}_k(\text{pic}(1), \text{pic}(2), \dots, \text{pic}(k)) \\ \text{pic}(k) &= \text{Fib}_k \left(2p(1) - \sum_{j=0}^1 p(j), 2p(2) - \sum_{j=0}^2 p(j), \dots, 2p(k) - \sum_{j=0}^k p(j) \right) \\ p(k) &= \sum_{j=0}^k 2^{\max(0, k-1-j)} \text{Fib}_j(\text{pic}(1), \text{pic}(2), \dots, \text{pic}(j)) \end{aligned}$$

These relationships can be naturally generalized by restricting the underlying arithmetic functions. Specifically, let $p_{>n}(k)$ denote the number of partitions of k into parts strictly greater than n , and let $\sigma_n(k)$ denote the restricted divisor sum function, representing the sum of divisors of k that are at most n . By adopting the first formula below as the formal definition of the generalized function $\text{pic}_n(k)$, all the preceding identities can be seamlessly mapped to this generalized setting as follows:

$$\begin{aligned}
\text{pic}_n(k) &:= Z_k(2^1 - \sigma_n(1) - 1, 2^2 - \sigma_n(2) - 1, \dots, 2^k - \sigma_n(k) - 1) \\
k \text{ pic}_n(k) &= \sum_{j=1}^k \text{pic}_n(k-j) (2^j - \sigma_n(j) - 1) \\
\text{pic}_n(k) &= \text{pic}_{n-1}(k) - \text{pic}_{n-1}(k-n) \\
\text{pic}_n\left(\binom{n+1}{2} + k\right) &= 2^{k-1} \left[2 \text{pic}_n\left(\binom{n+1}{2}\right) - (-1)^n \right] \\
\sigma_n(k) - 2^k + 1 &= \text{Fab}_k(\text{pic}_n(1), \text{pic}_n(2), \dots, \text{pic}_n(k)) \\
\sum_{j=0}^k p_{>n}(j) - 2p_{>n}(k) &= -\text{Fib}_k(\text{pic}_n(1), \text{pic}_n(2), \dots, \text{pic}_n(k)) \\
\text{pic}_n(k) &= \text{Fib}_k\left(2p_{>n}(1) - \sum_{j=0}^1 p_{>n}(j), 2p_{>n}(2) - \sum_{j=0}^2 p_{>n}(j), \dots, 2p_{>n}(k) - \sum_{j=0}^k p_{>n}(j)\right) \\
p_{>n}(k) &= \sum_{j=0}^k 2^{\max(0, k-1-j)} \text{Fib}_j(\text{pic}_n(1), \text{pic}_n(2), \dots, \text{pic}_n(j))
\end{aligned}$$

Furthermore, the following connection holds between the newly introduced generalized function $\text{pic}_n(k)$ and the classical sequence $\text{pic}(k)$:

$$\sum_{k=0}^n \text{pic}_{n-k}(k) = 2^n - \text{pic}(n+1) \quad (n = 0, 1, 2, \dots).$$

For all $n \geq 0$ let $\omega_n(k)$ (cf. OEIS:A231599 and A333290) defined by

$$\omega_n(k) := \begin{cases} 1 & \text{if } n, k = 0, \\ 0 & \text{if } n = 0 \text{ and } k > 0, \\ \text{the coefficient of } x^k \text{ in the expansion of the product } \prod_{j=1}^n (1 - x^j) & \text{if } n > 0, k = 0, 1, 2, \dots \end{cases}$$

For each $n \geq 0$, we introduce the following sequences:

$$\begin{aligned}
\omega_n &:= (\omega_n(0), \omega_n(1), \omega_n(2), \omega_n(3), \dots), & \mathbf{p} &:= (p(0), p(1), p(2), p(3), \dots), \\
\mathbf{pic} &:= (\text{pic}(0), \text{pic}(1), \text{pic}(2), \dots), & \mathbf{p}_{>n} &:= (p_{>n}(0), p_{>n}(1), p_{>n}(2), \dots), \\
\mathbf{pic}_n &:= (\text{pic}_n(0), \text{pic}_n(1), \text{pic}_n(2), \dots), & \mathbf{p}_{\leq n} &:= (p_{\leq n}(0), p_{\leq n}(1), p_{\leq n}(2), \dots),
\end{aligned}$$

where $p_{\leq n}(k)$ denotes the number of partitions of k into parts $\leq n$ (or equivalently, into at most n parts).

($\mathbf{p}_{>n}$: cf. OEIS:A000041, A002865, A008483, A008484, A185325-A185329)

($\mathbf{p}_{\leq n}$: cf. OEIS:A000012, A004526, A001399, A001400, A001401, A001402, A026813-A026816)

By definition, the following limiting cases immediately follow from the interpretations of these sequences:

$$\omega_0 = (1, 0, 0, 0, \dots), \omega_\infty = \omega, \mathbf{p} = \mathbf{p}_{>0} = \mathbf{p}_{\leq \infty}, \mathbf{0} = (0, 0, 0, \dots) = \mathbf{p}_{>\infty} = \mathbf{p}_{\leq 0}, \mathbf{1} = (1, 1, 1, \dots) = \mathbf{p}_{\leq 1}.$$

We will use the well-known inverse relation

$$\omega_n^{-*} = \mathbf{p}_{\leq n} \quad (n = 0, 1, 2, \dots, \infty)$$

where the superscript $\mathbf{a}^{-*}$ denotes the convolutional inverse of a sequence $\mathbf{a}$.

Using these algebraic formulations and sequence definitions, the relations established above for $\text{pic}_n(k)$ can be rewritten in a highly compact, equivalent form via the following convolutions:

$$\begin{aligned}
\mathbf{p}_{>n} &= \omega_n * \mathbf{p} \\
\mathbf{pic}_n &= \mathbf{p}_{>n} * \mathbf{pic}
\end{aligned}$$

To illustrate these algebraic properties, we explicitly compute the initial terms of the sequences $\mathbf{pic}_n$ corresponding to the cases $n = 0, 1$, and 2 .

$$\begin{aligned}
\mathbf{pic}_0 &= \mathbf{p}_{>0} * \mathbf{pic} = \mathbf{p} * \mathbf{pic} \\
&= (1, 1, 2, 3, 5, 7, 11, 15, 22, 30, 42, 56, 77, 101, \dots) * (1, 0, 0, 1, 2, 5, 9, 19, 37, 74, 148, 296, 591, 1183, \dots) \\
&= (1, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, \dots).
\end{aligned}$$

$$\begin{aligned}
\mathbf{pic}_1 &= \mathbf{p}_{>1} * \mathbf{pic} \\
&= (1, 0, 1, 1, 2, 2, 4, 4, 7, 8, 12, 14, 21, 24, \dots) * (1, 0, 0, 1, 2, 5, 9, 19, 37, 74, 148, 296, 591, 1183, \dots) \\
&= (1, 0, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, \dots).
\end{aligned}$$

$$\begin{aligned}
\mathbf{pic}_2 &= \mathbf{p}_{>2} * \mathbf{pic} \\
&= (1, 0, 0, 1, 1, 1, 2, 2, 3, 4, 5, 6, 9, 10, \dots) * (1, 0, 0, 1, 2, 5, 9, 19, 37, 74, 148, 296, 591, 1183, \dots) \\
&= (1, 0, 0, 2, 3, 6, 12, 24, 48, 96, 192, 384, 768, 1536, \dots).
\end{aligned}$$

The following matrix summarizes the first 14 terms of the sequences $\mathbf{pic}_n$ computed for $n = 0, 1, \dots, 9$.

$$\begin{bmatrix}
1 & 1 & 2 & 4 & 8 & 16 & 32 & 64 & 128 & 256 & 512 & 1024 & 2048 & 4096 & \dots \\
1 & 0 & 1 & 2 & 4 & 8 & 16 & 32 & 64 & 128 & 256 & 512 & 1024 & 2048 & \dots \\
1 & 0 & 0 & 2 & 3 & 6 & 12 & 24 & 48 & 96 & 192 & 384 & 768 & 1536 & \dots \\
1 & 0 & 0 & 1 & 3 & 6 & 10 & 21 & 42 & 84 & 168 & 336 & 672 & 1344 & \dots \\
1 & 0 & 0 & 1 & 2 & 5 & 10 & 20 & 38 & 76 & 152 & 305 & 610 & 1221 & \dots \\
1 & 0 & 0 & 1 & 2 & 5 & 9 & 20 & 38 & 75 & 150 & 300 & 600 & 1201 & \dots \\
1 & 0 & 0 & 1 & 2 & 5 & 9 & 19 & 38 & 75 & 149 & 298 & 595 & 1192 & \dots \\
1 & 0 & 0 & 1 & 2 & 5 & 9 & 19 & 37 & 75 & 149 & 297 & 593 & 1187 & \dots \\
1 & 0 & 0 & 1 & 2 & 5 & 9 & 19 & 37 & 74 & 149 & 297 & 592 & 1185 & \dots \\
1 & 0 & 0 & 1 & 2 & 5 & 9 & 19 & 37 & 74 & 148 & 297 & 592 & 1184 & \dots
\end{bmatrix}$$

By shifting each row of this matrix by its corresponding index $n = 0, 1, 2, 3, \dots$, we obtain precisely the matrix $\mathbf{R}$ discussed in Conjecture 1:

$$\mathbf{R} = \begin{bmatrix}
1 & 1 & 2 & 4 & 8 & 16 & 32 & 64 & 128 & 256 & 512 & 1024 & 2048 & 4096 & \dots \\
0 & 1 & 0 & 1 & 2 & 4 & 8 & 16 & 32 & 64 & 128 & 256 & 512 & 1024 & \dots \\
0 & 0 & 1 & 0 & 0 & 2 & 3 & 6 & 12 & 24 & 48 & 96 & 192 & 384 & \dots \\
0 & 0 & 0 & 1 & 0 & 0 & 1 & 3 & 6 & 10 & 21 & 42 & 84 & 168 & \dots \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & 6 & 10 & 20 & 39 & 78 & \dots \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & 5 & 10 & 20 & 38 & \dots \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & 5 & 9 & 20 & \dots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & 5 & 9 & \dots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & 5 & \dots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 2 & \dots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & \dots
\end{bmatrix}$$

By combining the two fundamental convolution formulas above, we immediately obtain the following identity for $\mathbf{pic}_n$:

$$\mathbf{pic}_n = \mathbf{p}_{>n} * \mathbf{pic} = \omega_n * (\mathbf{p} * \mathbf{pic}) = \omega_n * (1, 1, 2, 4, 8, 16, 32, \dots)$$

Convolving both sides of this equation with ω_n^{-*} , we arrive at:

$$\omega_n^{-*} * \mathbf{pic}_n = \mathbf{p}_{\leq n} * \mathbf{pic}_n = \mathbf{p} * \mathbf{pic} = (1, 1, 2, 4, 8, 16, 32, \dots)$$

Alternatively, convolving both sides with $\mathbf{p}^{-*} = \omega$ we get the remarkably symmetric relation:

$$\omega * \mathbf{pic}_n = \omega_n * \mathbf{pic}$$

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